



MY ROCK AND

MY FORTRESS

MATHEMATICS

YEAR 12

1999 HALF YEARLY EXAM

4 UNIT

*Time allowed: Three hours
(Plus 5 minutes reading time)*

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- ALL questions are of equal value.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Standard integrals are printed on page 10.
- Board-approved calculators may be used.
- Answer each question in a *separate* Writing Booklet.
- You may ask for extra Writing Booklets if you need them.

QUESTION 1

Marks

- (a) Find the modulus and argument of $Z = 3 + 4i$
(Express the argument in radians) 2
- (b) For any complex number Z where $Z = -\bar{Z}$ prove that Z must be purely imaginary 3
- (c) Find the square root of $Z = 5 - 12i$ 4
- (d) Draw a neat sketch to illustrate the following region of the Argand diagram 2
- $$-\frac{\pi}{6} \leq \arg(Z - 1) \leq \frac{\pi}{6} \text{ and } |Z - 1| \leq 1$$
- (e) If Z is a complex number such that 4
- $$|Z - 6| + |Z + 6| = 60$$
- describe geometrically the locus of Z and find its Cartesian equation.

Marks

QUESTION 2

- (a) If $1 + i$ is a solution of $x^4 - 6x^3 + 5x^2 + 2x - 10 = 0$ solve the equation over the field of real numbers. 3
- (b) If α, β, δ are the roots of $x^3 - px + q = 0$ find in terms of p and q a cubic equation with roots
- $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\delta}$
 - $\alpha^3, \beta^3, \delta^3$
- (c) If the cubic equation $2x^3 - 9x^2 + 12x + k = 0$ has two equal roots, find the value of k 4
- (d) Find the condition (i.e. the relationship between a and b) that $x^4 - 3ax + b = 0$ has a repeated root 4

QUESTION 3

Marks

- (a) Express z in the form $a + ib$ if

3

$$\arg(z+1) = \frac{\pi}{6} \text{ and } \arg(z-1) = \frac{2\pi}{3}$$

- (b) If z is a complex number show that $z^2 + (\bar{z})^2 = 2$ is a hyperbola and state its eccentricity

4

- (c) By writing each factor in the modulus-argument form, simplify

3

$$(\sqrt{3} + i)^6 \div (1 - i)^4$$

- (d) i. Find the four complex roots of $z^4 + 4 = 0$
ii. Plot these roots on an Argand diagram

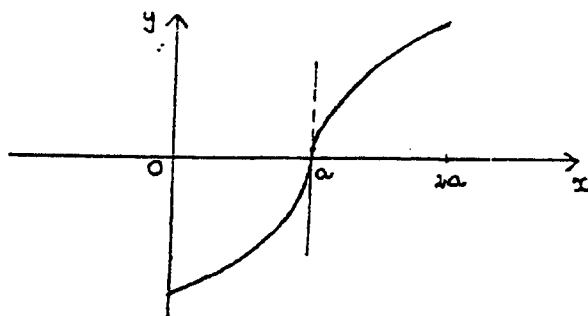
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QUESTION 4

Marks

- (a) Consider the graph of $y = f(x)$ for $0 \leq x \leq 2a$

3



The graph has point symmetry and a vertical tangent exists at $x = a$.
Sketch:

i. $y = f'(x)$

ii. $y = f''(x)$

iii. $y = \int_0^x f(t) dt$

- (b) i. Given $F(x) = \frac{x^2 - 1}{x^2 + 1}$, sketch the following on separate axes

12

1. $y = F(x)$

2. $[F(x)]^2 = \frac{x^2 - 1}{x^2 + 1}$

3. $y = [F(x)]^2$

4. $y = \log_e F(x)$

5. $y = \frac{x + 1[(x - 1)]}{x^2 + 1}$

- ii. Use your graph in (5) to solve the inequality $x^2 + 1 > 2[x + 1](x - 1)$

QUESTION 5

Marks

- (a) i. Obtain the equation of the tangent to the curve $\sqrt{x} + \sqrt{y} = \sqrt{c}$ at the point $P(a,b)$ on the curve 5
- ii. This tangent meets the x and y axes at Q and R respectively. Show that $OQ + OR = c$ for all positions of P , where O is the origin

- (b) i. Find the eccentricity, the equations of the directrices and the co-ordinates of the foci of the ellipse with equation $7x^2 + 16y^2 = 112$ 10
- ii. Sketch the ellipse showing the above information on your diagram. Also sketch the auxiliary circle on your diagram
- iii. Set up the integrals that give:
1. The area of a quadrant of the circle with equation $x^2 + y^2 = 16$
 2. The area of the quadrant of the ellipse $7x^2 + 16y^2 = 112$
- iv. Show that the integral in (iii) 2. above is $\frac{b}{a}$ times the integral in (iii) 1. and deduce the area of the ellipse from the known area of the circle. Hence write down a general formula for the area of an ellipse

QUESTION 6

Marks

- (a) The point $P(cp, \frac{c}{p})$ lies on the rectangular hyperbola $xy = c^2$ in the first quadrant. The tangent to the hyperbola at the point P, crosses the x axis at the point A and the y axis at the point B.

10

- i. Find the equation of the tangent to the hyperbola at the point P
- ii. Show that the equation of the normal to the hyperbola at the point P is $p^3x - py = cp^4 - c$
- iii. If the normal at P meets the other branch of the hyperbola at the point Q, determine the coordinates of Q
- iv. Show that the area of the triangle ABQ is $c^2 \left(p^2 + \frac{1}{p^2} \right)^2$
- v. Prove that the area of the triangle is a minimum when $p = 1$

- (b) If a, b and c are positive real numbers such that $a \neq b \neq c$, prove,

5

i. $\frac{a}{b} + \frac{b}{a} > 2$

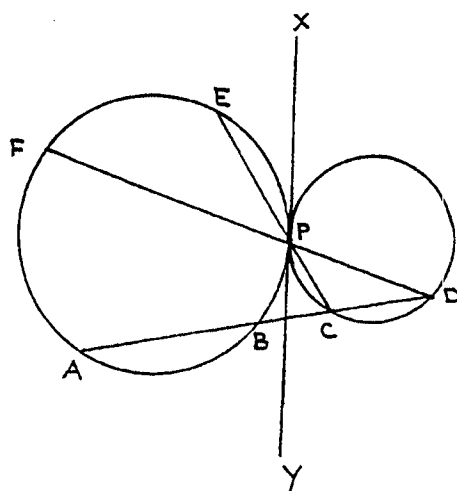
ii. $(a + b + c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) > 9$

QUESTION 7

Marks

- (a) Two circles touch externally at point P. The line ABCD cuts the first circle at A and B and the second circle at C and D. The lines CPE and DPF meet the first circle at E and F respectively. XPY is the common tangent.

7



Copy the diagram onto your answer paper.
Prove that:

- i. $FE \parallel AD$
- ii. $\angle FPA = \angle BPC$
- iii. $\triangle FPA \parallel \triangle BPC$

- (b) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is the equation of a hyperbola with eccentricity e .

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- i. Prove the perpendicular from the focus $S(ae, 0)$ to the asymptote $y = \frac{b}{a}x$ meets it on the directrix.
- ii. Prove that the angle between the two asymptotes is $2 \tan^{-1} \sqrt{e^2 - 1}$

QUESTION 8

Marks

- (a) Find, as a relation between k , l , and m , the condition for the quadratic equation in x ,

3

$(k^2 + l^2)x^2 + 2l(k + m)x + (l^2 + m^2) = 0$
to have real roots. Simplify your answer as far as possible.

- (b) If $|a| > 2|b|$, prove $2|a - b| > |a|$

3

- (c) i. Show that $\int_0^{\frac{\pi}{2}} \cos^4 x dx = \frac{3\pi}{16}$

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- ii. Prove $3(\cos^4 x + \sin^4 x) - 2(\cos^6 x + \sin^6 x) = 1$

- iii. Without attempting to evaluate any integrals, explain why:

$$\int_0^{\frac{\pi}{2}} \sin^n x dx = \int_0^{\frac{\pi}{2}} \cos^n x dx, \text{ for all positive integers } n$$

- iv By integrating the identity in part (ii), and using parts (i) and (iii),

$$\text{find } \int_0^{\frac{\pi}{2}} \cos^6 x dx$$

- v. Without attempting to evaluate any integrals, explain why:

$$\int_0^{\frac{\pi}{2}} \sin^{n+1} x dx < \int_0^{\frac{\pi}{2}} \sin^n x dx, \text{ for all positive integers } n$$

4 unit Solutions

1 a $3+4i = 5\left(\frac{3}{5} + \frac{4i}{5}\right)$

$\therefore \cos \theta = \frac{3}{5}$

$\theta = \cos^{-1} \frac{3}{5}$
 $= 92.7295218^\circ$
 $= 93^\circ$

$\therefore 3+4i = 5 \text{cis}(93)$

$\therefore \text{modulus} = 5, \text{ argument} = 0.93^\circ$

b) let $z = x+iy$

if $z = -\bar{z}$

$x+iy = -(x-iy)$

i.e. $2x = 0$

$x = 0$

Hence $z = 0+iy$

$= iy$

which is purely imaginary

c) let $x+iy = \sqrt{5-12i}$

i.e. $(x+iy)^2 = 5-12i$

$x^2 - y^2 + 2xyi = 5-12i$

Comparing real and imaginary parts

$x^2 - y^2 = 5 \dots 1)$

$2xy = -12 \dots 2)$

from 2) $y = \frac{-6}{x} \dots 3)$

Substitute 3) into 1)

$x^2 - \frac{36}{x^2} = 5$

$x^4 - 5x^2 - 36 = 0$

$(x^2+4)(x^2-9) = 0$

Hence $x = \pm 3$ (x is real)

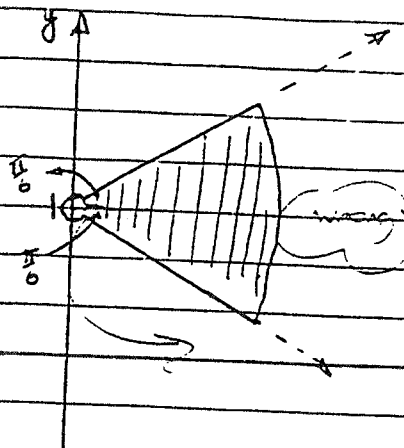
When $x = 3$ $y = -2$

$x = -3$ $y = 2$

$\therefore \sqrt{5-12i} = \pm(3-2i)$

d) $-\frac{\pi}{6} \leq \arg(z-1) \leq \frac{\pi}{6}$

and $|z-1| \leq 1$



e) $|z-6| + |z+6| = 60$

The locus of z is an ellipse, with foci at $(6,0)$ and $(-6,0)$. The length of the major axis is 60.
 i.e. $(a=30)$

$\therefore a = 30$

$ae = 6$

$e = \frac{1}{5}$

Now

$b^2 = a^2(1-e^2)$
 $= 900\left(1 - \frac{1}{25}\right)$

$= 864$

Hence the equation is

$\frac{x^2}{900} + \frac{y^2}{864} = 1$

Question 2.

a) Let $f(x) = x^4 - 6x^3 + 5x^2 + 2x - 10$

$f(x)$ has real coefficients

$\therefore$ if $1+i$ is a root $1-i$ is also
and ~~$(1-i)$~~

and $(x-1-i)(x-1+i)$ is a factor.

i.e. $x^2 - 2x + 2$ is a factor

$$x^2 - 4x - 5$$

$$\begin{array}{r} x^2 - 2x + 2 \end{array} \overline{) x^4 - 6x^3 + 5x^2 + 2x - 10}$$

$$x^4 - 2x^3 + 2x^2$$

$$-4x^3 + 3x^2 + 2x$$

$$-4x^3 + 8x^2 - 8x$$

$$-5x^2 + 10x - 10$$

$$-5x^2 + 10x - 10$$

$$0$$

$$\begin{aligned} \therefore f(x) &= (x^2 - 2x + 2)(x^2 - 4x - 5) \\ &= (x^2 - 2x + 2)(x+1)(x-5) \end{aligned}$$

Hence the real roots are $5, -1$

b) Let $f(x) = x^3 - px + q$

i) Put $u = \frac{1}{x} \therefore x = \frac{1}{u}$

Hence $\left(\frac{1}{u}\right)^3 - \frac{p}{u} + q = 0$ has roots
 $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$

$$\frac{1}{u^3} - \frac{p}{u} + q = 0$$

$$1 - pu^2 + qu^3 = 0$$

$$\text{i.e. } qu^3 - pu^2 + 1 = 0$$

So the required function is
 $qx^3 - px^2 + 1 = 0$

ii) Put $u = x^3 \Rightarrow x = u^{1/3}$

$$\left(u^{1/3}\right)^3 - pu^{1/3} + q = 0 \text{ has roots } \alpha^3, \beta^3, \gamma^3$$

$$u - pu^{1/3} + q = 0$$

$$u + q = pu^{1/3}$$

$$(u+q)^3 = (pu^{1/3})^3$$

$$u^3 + 3u^2q + 3uq^2 + q^3 = p^3u$$

$$u^3 + 3u^2q + (3q^2 - p^3)u + q^3 = 0$$

i.e.

$$x^3 + 3x^2q + (3q^2 - p^3)x + q^3 = 0$$

c) $f(x) = 2x^3 - 9x^2 + 12x + k = 0$
 $f'(x) = 6x^2 - 18x + 12$

If x is a double root then $f'(x) = 0$
and $f(x) = 0$

$$f'(x) = 0 \text{ when } 6(x^2 - 3x + 2) = 0$$

$$\text{i.e. } (x-1)(x-2) = 0$$

$$x = 1, 2$$

$$f(1) = 2 - 9 + 12 + k = 0 \Rightarrow k = -5$$

$$f(2) = 16 - 36 + 24 + k = 0 \Rightarrow k = -4$$

$\therefore$ the equation has equal roots
when $k = -4, -5$.

Question 2

$$a) \quad f(x) = x^4 - 3ax + b \quad \dots 1)$$

$$f'(x) = 4x^3 - 3a \quad \dots 2)$$

If there is a repeated root $f'(x) = f(x) = 0$

$$2) \times x \quad 4x^4 - 3ax = 0 \quad \dots 3)$$

$$x^4 - 3ax + b = 0 \quad \dots 1)$$

$$3) - 1) \quad 3x^4 - b = 0$$

$$x^4 = \frac{b}{3} \quad \dots 4)$$

divide $4x^3 - 3a = 0$

$$x^3 = \frac{3a}{4} \quad \dots 5)$$

from 4) $x^{12} = \left(\frac{b}{3}\right)^3 = \frac{b^3}{27}$

from 5) $x^{12} = \left(\frac{3a}{4}\right)^4 = \frac{81a^4}{256}$

$$\therefore \frac{b^3}{27} = \frac{81a^4}{256}$$

$$\text{i.e. } 256b^3 = 2187a^4$$

Question 3

$$a) \quad \arg(z-1) - \arg(z+1) = \frac{2\pi}{3} - \frac{\pi}{6}$$

$$\therefore \arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2}$$

i.e. the locus is the top semi-circle of $x^2 + y^2 = 1$ $x \neq \pm 1$

$$\text{i.e. } y = \sqrt{1-x^2} \quad x \neq \pm 1$$

$$\text{i.e. } |z| = 1$$

$$\text{Hence } z = x + (\sqrt{1-x^2})i$$

$$b) \quad \text{let } z = x+iy, \quad \bar{z} = x-iy$$

$$\text{hence } z^2 + \bar{z}^2 = 2$$

$\Rightarrow$

$$(x+iy)^2 + (x-iy)^2 = 2$$

$$x^2 + 2ixy + i^2y^2 + x^2 - 2ixy + i^2y^2 = 2$$

$$2x^2 - 2y^2 = 2$$

$$x^2 - y^2 = 1$$

which is a hyperbola

$$\text{Now } b^2 = a^2(e^2 - 1) \quad a=1, b=1$$

$$\therefore e^2 = 2$$

$$e = \sqrt{2}$$

so we have a hyperbola with eccentricity $\sqrt{2}$

$$c) \quad \text{let } z_1 = \sqrt{3} + i \quad |z_1| = \sqrt{3+1} = 2$$

$$\arg z_1 = \tan^{-1} \frac{1}{\sqrt{3}} = \frac{\pi}{6}$$

$$\therefore z_1 = 2 \operatorname{cis} \frac{\pi}{6}$$

$$z_2 = 1-i \quad |z_2| = \sqrt{1+1} = \sqrt{2}$$

$$\arg z_2 = -\tan^{-1} 1 = -\frac{\pi}{4}$$

$$\therefore z_2 = \sqrt{2} \operatorname{cis} \left(-\frac{\pi}{4}\right)$$

$$(\sqrt{3}+i)^6 \div (1-i)^4 = \left(2 \operatorname{cis} \frac{\pi}{6}\right)^6 \div \left(\sqrt{2} \operatorname{cis} \left(-\frac{\pi}{4}\right)\right)^4$$

$$= 2^6 \operatorname{cis} \pi \div 2^2 \operatorname{cis} (-\pi)$$

$$= 2^4 \operatorname{cis} (\pi - (-\pi))$$

$$= 2^4 \operatorname{cis} 2\pi$$

$$= 2^4 \cos 2\pi + i \sin 2\pi$$

$$= 2^4$$

$$= 16$$

$$d) z^4 + 4 = 0$$

$$i) z^4 = -4$$

$$\therefore r^4 (\cos \theta + i \sin \theta)^4 = -4$$

$$r^4 (\cos 4\theta + i \sin 4\theta) = -4$$

$$\therefore r^4 = 4$$

$$\cos 4\theta + i \sin 4\theta = -1$$

$$r = \sqrt{2}$$

$$\cos 4\theta = -1 \quad \text{and} \quad \sin 4\theta = 0$$

$$4\theta = \pi, 3\pi, 5\pi, 7\pi$$

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

Roots

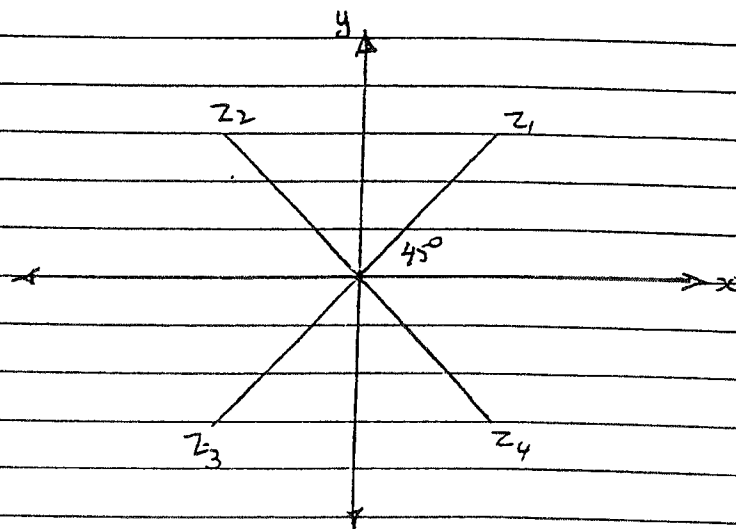
$$z_1 = \sqrt{2} \cos \frac{\pi}{4}$$

$$z_2 = \sqrt{2} \cos \frac{3\pi}{4}$$

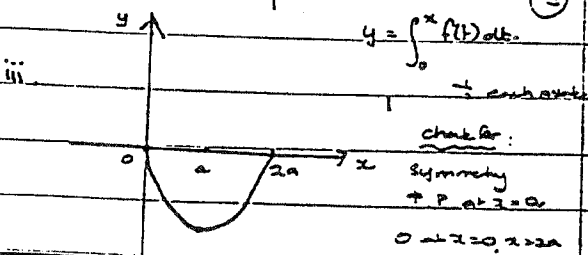
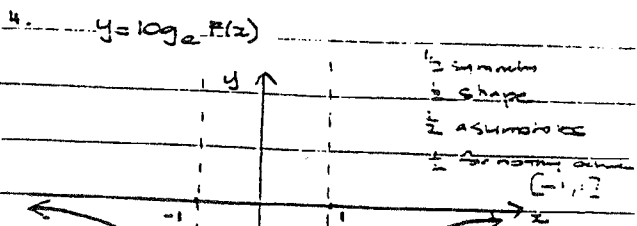
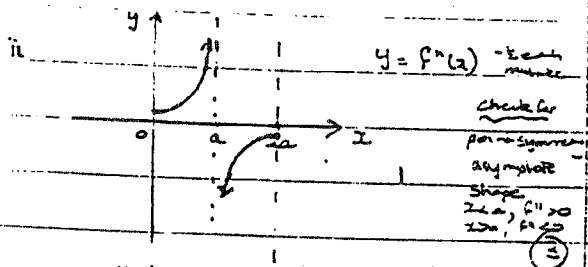
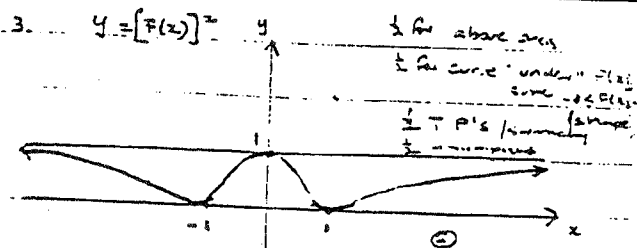
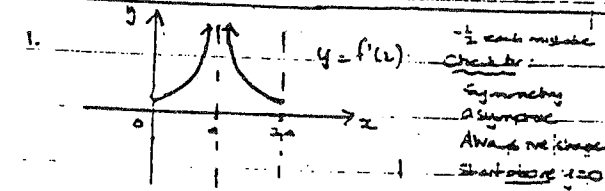
$$z_3 = \sqrt{2} \cos \frac{5\pi}{4}$$

$$z_4 = \sqrt{2} \cos \frac{7\pi}{4}$$

ii)



④(a)



5. $y = \frac{|x+1|(x-1)}{x^2+1}$

(b) i. $F(x) = \frac{x^2-1}{x^2+1}$

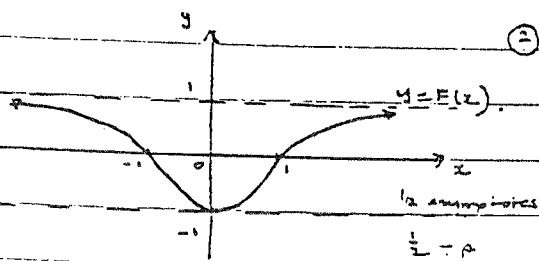
1. even function, $F(x) = 1 - \frac{2}{x^2+1}$

$F(x) = 0, x = \pm 1; x = 0, F(x) = -1$ (min. value)

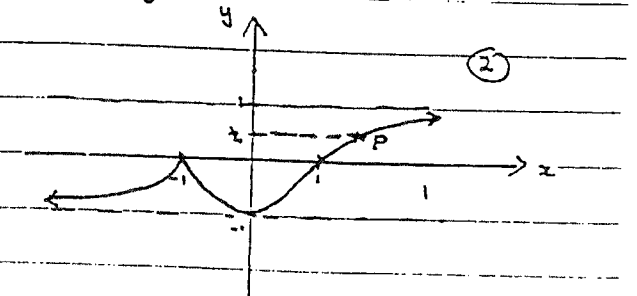
$-1 \leq F(x) < 1$

As $x \rightarrow \infty, F(x) \rightarrow 1$

As $x \rightarrow -\infty, F(x) \rightarrow 1$



$= \begin{cases} F(x) & \text{if } x \geq -1 \\ -F(x) & \text{if } x < -1 \end{cases}$



(b) $x^2+1 > 2|x+1|(x-1)$

$\frac{|x+1|(x-1)}{x^2+1} < \frac{1}{2}$

To find P,

Solve $F(x) = \frac{1}{2}$ (since $x > -1$)

$2(x^2-1) = x^2+1$

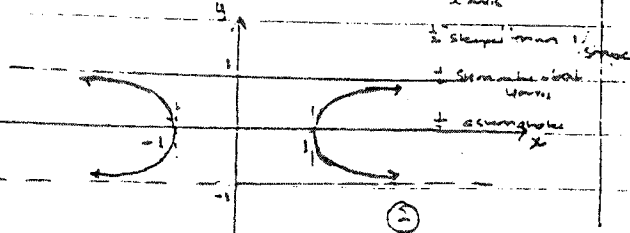
$x^2-3=0$

$x = \sqrt{3} \quad (x > 0)$

$x^2+1 > 2|x+1|(x-1)$ for all $x < \sqrt{3}$

(From graph)

2. $[F(x)]^2 = \frac{x^2-1}{x^2+1}$



(5) (i) $\sqrt{x} + \sqrt{y} = \sqrt{c}$

Diff. w.r.t x : $\frac{1}{2}x^{-\frac{1}{2}} + \frac{1}{2}y^{-\frac{1}{2}} \frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\frac{\sqrt{y}}{\sqrt{x}}$

$\therefore$ At $P(a, b)$, $\frac{dy}{dx} = -\sqrt{\frac{b}{a}}$

$\therefore$ Eqn. of tangent is

$y - b = -\sqrt{\frac{b}{a}}(x - a)$

$\sqrt{a}y - b\sqrt{a} = -\sqrt{b}x + a\sqrt{b}$

i.e. $\sqrt{b}x + \sqrt{a}y = a\sqrt{b} + b\sqrt{a}$ (5)

(ii) when $y=0$, $x = \frac{a\sqrt{b} + b\sqrt{a}}{\sqrt{b}}$

$= a + \sqrt{ab} \therefore Q(a + \sqrt{ab}, 0)$

$x=0$, $y = \frac{a\sqrt{b} + b\sqrt{a}}{\sqrt{a}}$

$= \sqrt{ab} + b \therefore R(0, b + \sqrt{ab})$

$\therefore OQ = a + \sqrt{ab}$ units

$OR = b + \sqrt{ab}$ units

$OQ + OR = a + b + 2\sqrt{ab}$

$= (\sqrt{a} + \sqrt{b})^2$

$= (\sqrt{c})^2 = c$ (since a, b satisfy eqn of curve)

(b) i) $7x^2 + 16y^2 = 112$

i.e. $\frac{x^2}{16} + \frac{y^2}{7} = 1 \therefore a=4, b=\sqrt{7}$

$b^2 = a^2(1 - e^2)$

$7 = 16(1 - e^2)$

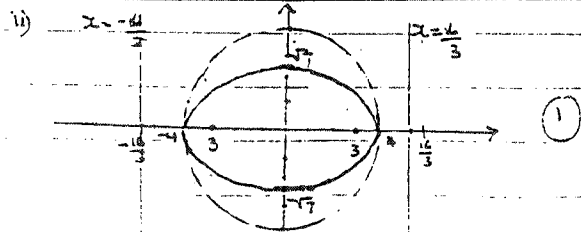
$e^2 = 1 - \frac{7}{16} = \frac{9}{16}$

$\therefore e = \frac{3}{4}$ ($e > 0$)

$\therefore$ eccentricity = $\frac{3}{4}$

Directrices are $x = \pm \frac{16}{3}$

Foci are $(3, 0)$ and $(-3, 0)$



(iii) 1. $\int_0^4 \sqrt{16-x^2} dx$

2. $\int_0^4 \sqrt{\frac{112-7x^2}{16}} dx$

$= \frac{1}{4} \int_0^4 \sqrt{112-7x^2} dx$

(iv) $\frac{1}{4} \int_0^4 \sqrt{112-7x^2} dx$

$= \frac{1}{4} \int_0^4 \sqrt{7} \cdot \sqrt{16-x^2} dx$

$= \frac{\sqrt{7}}{4} \int_0^4 \sqrt{16-x^2} dx$

$= \frac{b}{a} \int_0^4 \sqrt{16-x^2} dx$

$\therefore$ Area of ellipse = $\frac{\sqrt{7}}{4} \times \frac{\pi r^2}{4}$ where $r=4$

$= \frac{\sqrt{7}}{4} \times 16\pi$

$= 4\sqrt{7}\pi$ units²

$\therefore$ General form for area of ellipse is

$A = \frac{b}{a} \times \pi a^2$

$= \pi ab$ units²

(6) (i) $xy = c^2$

$\frac{dy}{dx} = -\frac{c^2}{x^2}$

$\therefore$ At P , $m = -\frac{1}{p^2}$

$\therefore$ Eqn of tangent at P is

$y - \frac{c}{p} = -\frac{1}{p^2}(x - cp)$

$p^2y - cp = -x + cp$ (2)

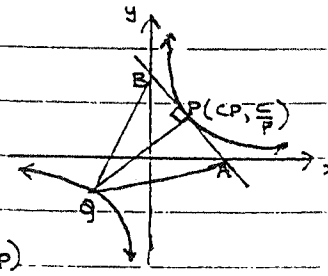
i.e. $x + p^2y = 2cp$

(ii) Eqn of normal is

$y - \frac{c}{p} = p^2(x - cp)$

$py - c = p^2x - cp^4$ (1)

i.e. $p^2x - py = cp^4 - c$



Alternative, p must be of form $(\frac{c}{p^3}, \frac{c}{p^3})$

(iii) Solve $xy = c^2$ — (1)

$p^3 x - py = cp^4 - c$ — (2) Simultaneously.

From (1) $y = \frac{c^2}{x}$, sub into (2)

$p^3 x - \frac{pc^2}{x} = cp^4 - c$

$p^3 x^2 - pc^2 = (cp^4 - c)x$

$p^3 x^2 - (cp^4 - c)x - pc^2 = 0$

We know $x = cp$ is one solution. Therefore:

$(x - cp)(p^3 x + c) = 0$

$\therefore x = cp$ or $x = -\frac{c}{p^3}$

$x = cp$ corr. to 1st quad. (2)

$x = -\frac{c}{p^3}$ is the x value at Q

Sub into (2)

$y = \frac{c^2 x}{x^2} = \frac{p^3}{c}$
 $= -cp^3$

$Q(-\frac{c}{p^3}, -cp^3)$

(iv) Now $A(2cp, 0)$ and $B(0, \frac{2c}{p})$

$AB = \sqrt{4c^2 p^2 + \frac{4c^2}{p^2}}$

$= \frac{2c}{p} \sqrt{p^4 + 1}$

$APB = \sqrt{(\frac{cp + \frac{c}{p^3}}{p^3})^2 + (\frac{c}{p} + cp^3)^2}$ (3)

$= \frac{c}{p^3} \sqrt{(p^4 + 1)^2 + p^2(1 + p^4)^2}$

$= \frac{c(p^4 + 1)}{p^3} \sqrt{1 + p^4}$

$\therefore \text{Area of } \triangle ABQ = \frac{1}{2} AB \cdot PB$

$= \frac{1}{2} \times \frac{2c}{p} \sqrt{p^4 + 1} \times \frac{c(p^4 + 1)}{p^3} \sqrt{p^4 + 1}$

$= \frac{c^2}{p^4} (p^4 + 1)^2$

$= c^2 (p^2 + \frac{1}{p^2})^2$

(v) $A = c^2 (p^2 + \frac{1}{p^2})^2$

$= c^2 [p^4 + 2 + \frac{1}{p^4}]$

$\frac{dA}{dp} = c^2 (4p^3 - \frac{4}{p^5})$

For min., $\frac{dA}{dp} = 0$

$4p^3 = \frac{4}{p^5}$

$p^8 = 1$

$\therefore p = 1$

$\frac{d^2 A}{dp^2} = 4c^2 [3p^2 + \frac{5}{p^6}]$
 $= 4c^2 [8]$

> 0 when $p = 1$

$\therefore p = 1$ corresponds to a minimum.

[i.e. when P is at the point (c, c)]

(b)

(b)(i) To prove: $\frac{a}{b} + \frac{b}{a} > 2$ $a \neq b$

a, b real pos. no.

Proof: $(\sqrt{\frac{a}{b}} - \sqrt{\frac{b}{a}})^2 > 0$ $a, b > 0$

$\therefore \frac{a}{b} + \frac{b}{a} - 2\sqrt{\frac{a}{b} \cdot \frac{b}{a}} > 0$

(2)

$\therefore \frac{a}{b} + \frac{b}{a} > 2$

(ii) To prove: $(a+b+c)(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}) > 9$

Proof: $(a+b+c)(\frac{1}{a} + \frac{1}{b} + \frac{1}{c})$

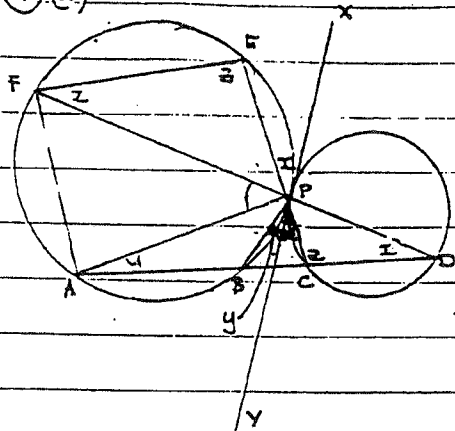
$= 1 + 1 + 1 + \frac{a}{b} + \frac{c}{a} + \frac{b}{c} + \frac{c}{b} + \frac{a}{c} + \frac{b}{a}$

$= 3 + \frac{a}{b} + \frac{b}{a} + \frac{a}{c} + \frac{c}{a} + \frac{b}{c} + \frac{c}{b}$

$> 3 + 2 + 2 + 2 = 9$

(from (i), $\frac{a}{b} + \frac{b}{a} > 2$ etc.)

⑦ (a)



(i) To Prove: $FE \parallel AD$

Proof: Let $\angle EPX = x$

then $\angle PFE = x$ (angle bet. chord & tangent equals to alt. segment) $\frac{1}{2}$

Also $\angle YPC = x$ (vertically opp. $\angle$'s) $\frac{1}{2}$

$\therefore \angle CDP = x$ (4 bet. chord & tangent equals to alt. segment) $\frac{1}{2}$

ie: $\angle FEP = \angle CDP$

$\therefore FE \parallel AD$ (alt. $\angle$'s equal) $\frac{1}{2}$

(ii) To prove: $\angle FPA = \angle BPC$

Proof: Let $\angle BPY = y$

then $\angle BPC = x + y$

Let $\angle FEP = z$

then $\angle DCP = z$ (alt. $\angle$'s, $FE \parallel AD$) $\frac{1}{2}$

$\therefore \angle PBC = z - (x + y)$ (ext. $\angle$ of $\triangle PBC$) $\frac{1}{2}$

Also, $\angle PAB = y$ (4 bet. chord & tangent equals to alt. segment) $\frac{1}{2}$

$\therefore \angle BPA = z - (x + y) - y$ (ext. $\angle$ of $\triangle PBA$)
 $= z - x - 2y$ $\frac{1}{2}$

And $\angle EPF = 180^\circ - (x + z)$ (4 sides of $\triangle FEP$) $\frac{1}{2}$

$\therefore \angle FPA = 180^\circ - [180^\circ - (x + z)] - [z - x - 2y]$
 $= [x + y]$ (EP, C is straight line) $\frac{1}{2}$

$= 180 - 180 + x + z - z + x + 2y = x + y$

$= x + y = \angle BPC$

(iii) To prove: $\triangle FPA \parallel \triangle BPC$

Proof: In $\triangle FPA$ and $\triangle BPC$,

$\angle FPA = \angle BPC$ (from (i) above) $\frac{1}{2}$

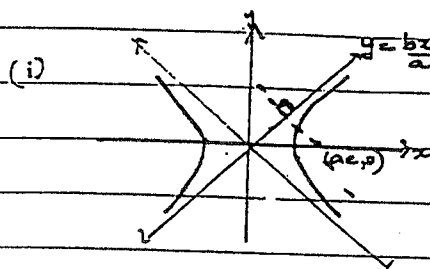
$\angle FAP = 180^\circ - z$ (opp. $\angle$'s of a cyclic quad. supp.) $\frac{1}{2}$

And $\angle BCP = 180^\circ - z$ (A, B, C, D straight line) $\frac{1}{2}$

$\therefore \angle FAP = \angle BCP$

$\therefore \triangle FPA \parallel \triangle BPC$ (equiangular) $\frac{1}{2}$

(b) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$



Line $\perp$ to $y = \frac{bx}{a}$ has gradient $-\frac{a}{b}$

and perp. through focus is

$$y - 0 = -\frac{a}{b}(x - ae)$$

$$\text{by } y = -ax + a^2e$$

$$\text{ie } ax + by = a^2e \quad 1$$

This meets the line $y = \frac{bx}{a}$ when

$$ax + b\left(\frac{bx}{a}\right) = a^2e$$

$$a^2x + b^2x = a^3e$$

$$x = \frac{a^3e}{a^2 + b^2} \quad 1$$

$$\text{But } b^2 = a^2(e^2 - 1)$$

$$\text{so } a^2 + b^2 = a^2e^2$$

$$\therefore x = \frac{a^3e}{a^2e^2} = \frac{a}{e} \quad \text{which is a pt. on the directrix}$$

$\therefore$ the perp. meets the asymptote on the directrix

(ii) Angle between the asymptotes is

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \text{ where } m_1 = \frac{b}{a}, m_2 = -\frac{b}{a}$$

$\therefore$ if $t = \tan \frac{\theta}{2}$,

$$\frac{2t}{1-t^2} = \frac{\frac{b}{a} + \frac{b}{a}}{1 - \frac{b^2}{a^2}}$$

$$\therefore \frac{2t}{1-t^2} = \frac{2(\frac{b}{a})}{1 - (\frac{b}{a})^2}$$

$$\therefore t = \frac{b}{a}$$

(Since $f(t) = \frac{2t}{1-t^2}$ is a function

only one value of $f(t)$ for each t .)

But $b^2 = a^2(e^2 - 1)$

$$\therefore \frac{b}{a} = \sqrt{e^2 - 1} \quad (\frac{b}{a} > 0)$$

(5)

$$\therefore \tan \frac{\theta}{2} = \sqrt{e^2 - 1}$$

$$\therefore \frac{\theta}{2} = \tan^{-1} \sqrt{e^2 - 1}$$

$$\therefore \theta = 2 \tan^{-1} \sqrt{e^2 - 1}$$

8. (a) $(k^2 + l^2)x^2 + 2l(k+m)x + (l^2 + m^2) = 0$

to have real roots, $\therefore \Delta \geq 0$

$$\Delta = 4l^2(k+m)^2 - 4(l^2 + m^2)(k^2 + l^2)$$

$$= 4l^2(k^2 + 2km + m^2) - 4(l^2k^2 + l^4 + m^2k^2 + m^2l^2)$$

$$= 4l^2k^2 + 8l^2km + 4l^2m^2 - 4l^2k^2 - 4l^4 - 4m^2k^2 - 4m^2l^2$$

$$= 8l^2km - 4l^4 - 4m^2k^2 - 4m^2l^2$$

$$= -4(l^4 - 2l^2km + m^2k^2)$$

$$= -4(l^2 - mk)^2$$

$$= -4(l^2 - mk)^2$$

$$\text{If } \Delta \geq 0, -4(l^2 - mk)^2 \geq 0$$

$$\therefore (l^2 - mk)^2 \leq 0$$

$$\text{But } (l^2 - mk)^2 \geq 0 \text{ as it is a square. } \therefore \text{Only possible value is}$$

$l^2 - mk = 0$ i.e. $l^2 = mk$

(b) To prove: $2|a-b| > |a|$ if $|a| > 2|b|$

Proof: $2|a-b| = 2|a(1 - \frac{b}{a})|$

$$= 2|a| |1 - \frac{b}{a}|$$

$$> 2|a| [1 - |\frac{b}{a}|]$$

(Since $|a-b| > |a| - |b|$)

$$\therefore 2|a-b| > 2|a| [1 - \frac{1}{2}]$$

(Since $|\frac{b}{a}| < \frac{1}{2}$)

$$= |a|$$

$$\therefore 2|a-b| > |a|$$

(c) i) $\cos^4 x = \frac{\cos 2x + 1}{2}$

$$\cos^4 x = \frac{1}{4} (1 + \cos 2x)^2$$

$$= \frac{1}{4} (1 + 2\cos 2x + \cos^2 2x)$$

$$= \frac{1}{4} [1 + 2\cos 2x + \frac{\cos 4x + 1}{2}]$$

$$\therefore \int_0^{\frac{\pi}{2}} \cos^4 x dx = \frac{1}{4} \int_0^{\frac{\pi}{2}} (\frac{3}{2} + 2\cos 2x + \frac{\cos 4x}{2}) dx$$

$$= [\frac{3x}{8}]_0^{\frac{\pi}{2}} + 0 + 0$$

these graphs are symmetrical about the x-axis for $0 < x < \frac{\pi}{2}$

$$= \frac{3\pi}{16}$$

ii) To prove:

$$3(\cos^4 x + \sin^4 x) - 2(\cos^6 x + \sin^6 x) = 1$$

$$\text{L.S.} = \cos^4 x + \sin^4 x + 2\cos^4 x + 2\sin^4 x$$

$$- 2\cos^6 x - 2\sin^6 x$$

$$= \cos^4 x + \sin^4 x + 2\cos^4 x(1 - \cos^2 x)$$

$$+ 2\sin^4 x(1 - \sin^2 x)$$

$$= \cos^4 x + \sin^4 x + 2\cos^4 x \sin^2 x$$

$$+ 2\sin^4 x \cos^2 x$$

$$= \cos^4 x + \sin^4 x + 2\sin^2 x \cos^2 x (\cos^2 x + \sin^2 x)$$

$$= \cos^4 x + \sin^4 x + 2\sin^2 x \cos^2 x$$

$$= (\cos^2 x + \sin^2 x)^2 = 1 = \text{R.S.}$$

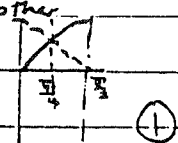
$0 < x < \pi$

(iii) $\cos(\frac{\pi}{2} - x) = \sin x$ $y = \cos x$ and

$y = \sin x$ are reflections of each other

in the line $x = \frac{\pi}{4}$

$$\therefore \int_0^{\frac{\pi}{2}} \cos x \, dx = \int_0^{\frac{\pi}{2}} \sin x \, dx$$



Similarly $y = \cos^2 x$ and $y = \sin^2 x$ must be

reflections of each other in the line $y = \frac{\pi}{4}$

$$\therefore \int_0^{\frac{\pi}{2}} \cos^2 x \, dx = \int_0^{\frac{\pi}{2}} \sin^2 x \, dx$$

(iv)

$$3 \int_0^{\frac{\pi}{2}} (\cos^4 x + \sin^4 x) \, dx - 2 \int_0^{\frac{\pi}{2}} (\cos^6 x + \sin^6 x) \, dx = \frac{\pi}{2}$$

But $\int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \int_0^{\frac{\pi}{2}} \cos^4 x \, dx$ and
similarly for $\int_0^{\frac{\pi}{2}} \sin^6 x \, dx$

$$\therefore 3 \int_0^{\frac{\pi}{2}} 2 \cos^4 x \, dx - 2 \int_0^{\frac{\pi}{2}} 2 \cos^6 x \, dx = \frac{\pi}{2}$$

From (i)

$$6 \left(\frac{3\pi}{16} \right) - 4 \int_0^{\frac{\pi}{2}} \cos^6 x \, dx = \frac{\pi}{2}$$

$$\therefore 4 \int_0^{\frac{\pi}{2}} \cos^6 x \, dx = \frac{9\pi}{8} - \frac{\pi}{2} \quad (2)$$

$$= \frac{5\pi}{8}$$

$$\therefore \int_0^{\frac{\pi}{2}} \cos^6 x \, dx = \frac{5\pi}{32}$$

(v) Since $0 \leq \sin x \leq 1$ for $0 \leq x \leq \frac{\pi}{2}$

then $\sin^n x \times \sin x \leq \sin^n x$

ie $\sin^{n+1} x \leq \sin^n x$ for all x (1)

$$0 < x < \frac{\pi}{2}$$

(for all the integers n).

$$\therefore \int_0^{\frac{\pi}{2}} \sin^{n+1} x \, dx < \int_0^{\frac{\pi}{2}} \sin^n x \, dx$$

for all the integers n .